

Deploy IoT-enabled smart grid management for cold ironing and shore power in maritime ports to optimize renewable energy usage and reduce vessel idle emissions.

Transportation and Logistics × Energy optimisation × IoT platform · OS070 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
23 is the world moving	50 can we play, can we win	HIGH 0 named in the evidence	€2.4bn observed evidence · per year	LATER research and standards activi...	L0 2 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying IoT-enabled smart grid management for cold ironing and shore power in maritime ports, helping ports optimize renewable energy usage and reduce vessel idle emissions. It targets port authorities and terminal operators who are under pressure to electrify operations and integrate green energy. The time is right because ports are actively transforming through electrification, IoT, 5G, and green energy, as evidenced by recent research.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€3.8bn €1.2bn – €18.2bn	€2.4bn €734m – €11.3bn	€35.5m €11.0m – €170m	observed
Observed: tendered contract value, annualised	€115m	€115m	€1.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting IoT platform	32.5 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

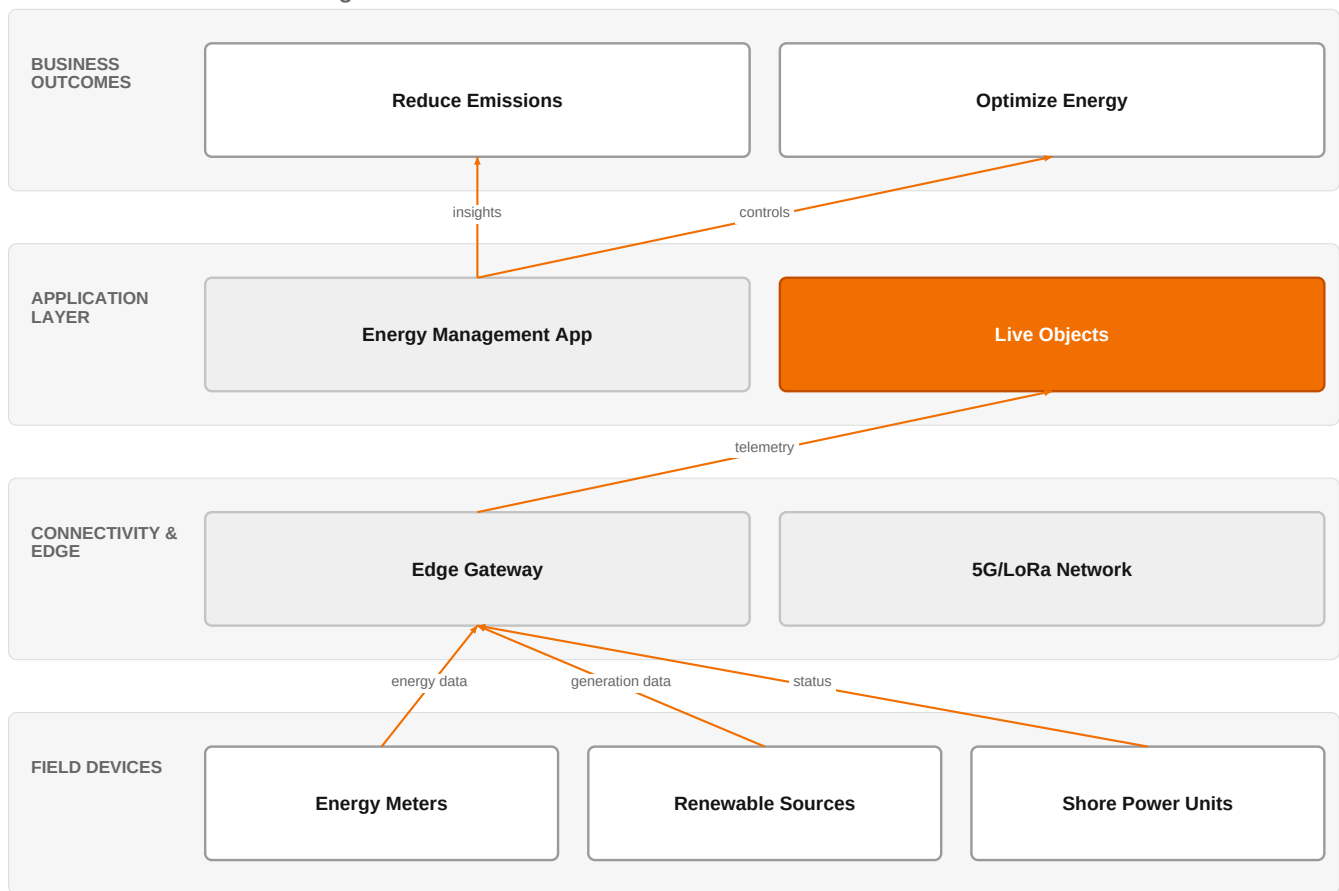
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only.

The solution, in one picture

Smart Grid Shore Power Management



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Live Objects platform connects field devices to an energy management application, enabling real-time optimization of shore power and renewable energy usage.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Maritime ports are undergoing a transformation driven by electrification, IoT, 5G, and green energy. Renewable energy sources are being integrated into power distribution systems, offering environmental benefits, cost-effective power generation, and improved energy sustainability. This shift is reducing reliance on fossil fuels and increasing energy efficiency. Additionally, IoT deployments are growing exponentially, but traditional cloud-based solutions face latency and energy overhead challenges, prompting a move toward edge AI for real-time fault detection and energy management.

Sources: SIG-F98C940013AB openalex.org 2025-08-22; SIG-F48074B55B20 openalex.org 2025-10-16

Who buys, and why now

The primary buyer is the port authority's head of infrastructure or energy manager, who feels the pain of rising energy costs and regulatory pressure to reduce emissions. The trigger is often a new environmental regulation or a sustainability target that forces the port to electrify its quayside operations. Operational pain includes managing intermittent renewable energy sources, ensuring reliable power supply for vessels at berth, and reducing idle emissions from auxiliary engines.

Why it is hot now — every claim bound to a dated source

- Renewable energy sources are being integrated into maritime systems like ports, with benefits including environmental friendliness, cost-effective power generation, and energy sustainability. [SIG-F98C940013AB]
- Renewable energy sources are being integrated into maritime systems to increase energy efficiency and reduce reliance on fossil fuels. [SIG-F98C940013AB]
- The transformation of maritime ports involves electrification, IoT, 5G, and green energy. [SIG-F98C940013AB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver an IoT platform (Live Objects) to connect and manage sensors and devices across the port's shore power infrastructure. The engagement would start with a discovery phase using Orange's IoT experts and Digital experts to assess the port's current energy infrastructure and identify integration points. Then, Live Objects would be deployed to collect data from energy meters, renewable generation units, and vessel connections, enabling real-time monitoring and control. Orange Group researchers could contribute to optimizing energy distribution algorithms, while Defense and security experts would ensure the solution meets sovereign and security requirements. The solution would be delivered in phases, starting with a pilot on one berth, then scaling across the port.

Why Orange can win

Orange can win because it combines a proven IoT platform (Live Objects) with deep expertise in IoT, digital, and security, and access to Orange Group researchers. The ability to deliver a sovereign, secure solution is a differentiator in the maritime sector, where critical infrastructure protection is paramount. However, the assets are relatively thin for this specific use case, as there is no dedicated energy management solution or maritime-specific reference, so Orange must leverage its platform and expertise to build a compelling value proposition.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; active in Transportation and Logistics; competes in Connectivity Solutions, OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current shore power infrastructure and how many berths are equipped?
- 2 Do you have renewable energy sources (solar, wind) integrated into your port's power grid?
- 3 What are your main operational challenges with managing energy demand and emissions from vessels at berth?
- 4 Are you under regulatory pressure to reduce emissions or meet specific sustainability targets?
- 5 How do you currently monitor and control energy usage across the port?
- 6 What is your timeline for electrification or smart grid initiatives?

Objections you will meet

They will say	You can answer
We already have an IoT platform from a hyperscaler.	That's true, and hyperscalers offer broad capabilities. However, our solution is designed for sovereign and secure environments, which is critical for port infrastructure. We can also integrate with your existing systems and provide specialized expertise in energy optimization.
This is too complex and expensive.	We understand that ports have tight budgets. Our phased approach starts with a pilot on one berth to demonstrate value and ROI before scaling. We also leverage our IoT platform to reduce integration costs.
We don't have the in-house expertise to manage this.	That's exactly why we bring our IoT and digital experts to the table. We handle the deployment, integration, and ongoing management, so your team can focus on core port operations.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-F98C940013AB	2025-08-22	openalex.org	1	Smart, Connected, and Sustainable: The Transformation of Maritime Ports Through Electrification...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:41+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.